

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 5

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Topic

Adapted Coordinates and Uniqueness

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Adapted Coordinates and Uniqueness. State all assumptions and conventions.

$$B_\mu(g, \partial g) = 0$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$T_a[g(x)] = T_a[g'(x')](a \in \text{Registry})$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$\mathcal{E}_{C5'}(c_1, \dots, c_n) = 0$$

Show enough intermediate work that another student can reproduce the calculation.